

7.1 Introduction to exploratory data analysis ↕

□ Activity: How many humans have ever lived?

Read the following webpage. As you read, take time to consider and make notes on:

- how accurate the data used for this visualisation is
- what is good about the visualisation and what is not as good.

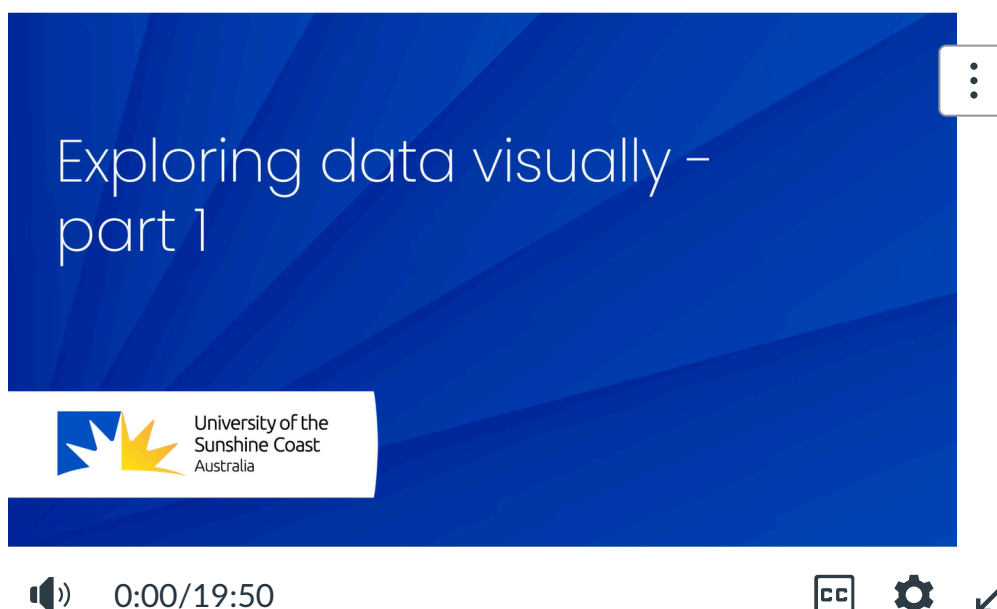
Explain your reasoning and bring your notes to the workshop.

Our World in Data, & Routley, N. (2022, March 25). [How many humans have ever lived?](https://www.visualcapitalist.com/cp/how-many-humans-have-ever-lived/) ↗ (<https://www.visualcapitalist.com/cp/how-many-humans-have-ever-lived/>)
Visual Capitalist.

□ Watch: Exploring data visually (part 1) [19.50 mins]

Watch the following video, which explains how to organise and arrange data to facilitate data exploration and visualisation. How to create and interpret visualisations to explore the distributions of individual variables is also described.

This is useful as it provides an understanding of how to conduct exploratory data analysis and how to use MS Excel tools to rearrange data to enable visual exploration of a dataset.



[Transcript](#)

[\(https://learn.usc.edu.au/courses/35242/files/3138648?wrap=1\)](https://learn.usc.edu.au/courses/35242/files/3138648?wrap=1)

Source: © University of the Sunshine Coast (2023).

Download the following slides for note taking: [ICT702 Exploring data visually \(part 1\) – exploratory analysis \(https://learn.usc.edu.au/courses/35242/files/3168009?wrap=1\)](https://learn.usc.edu.au/courses/35242/files/3168009?wrap=1)

The first step in analysing data is to examine and explore the data.

Understanding the data is gained through using descriptive statistics and visualisations. The objectives of the process are to

- detect errors, missing values and unusual observations
- gain understanding of the distribution of the values for individual variables
- identify patterns and relationships between the variables.

Creating visualisations is an essential part of the process. Visualisations enable the translation of data into information, providing an initial look at the data, which can fulfil the above objectives.

It is important to understand the structural dimensions of data:

- *Tall data* refers to there being a large number of records/rows in a table.
- *Wide data* refers to there being a large number of variables/columns in a table.

The taller and/or wider a table of data becomes, the greater the possibility of it containing erroneous or missing values. Wide data is more difficult to explore, as there are a greater number of possible variable combinations to examine.

Missing data can be visualised by adding *conditional formatting* to the data table, which adds colour to empty cells. The frequency and pattern of the missing values can then be identified visually.

Most spreadsheet software also provides the functionality to *filter* and *sort* data. Data can be sorted:

- alphabetically, for categorical data
- chronologically, by date
- according to magnitude, for quantitative data.

□ Watch: Importance of exploratory data analysis [3.32 mins]

This video discusses the consequences of data errors, introduces exploratory data analysis and explains why it is essential.

As you watch, consider:

- why exploratory data analysis may improve the final data visualisations that are created
- how a deeper understanding of a dataset can result in a better audience experience.



Source: © Precision Health (2021).